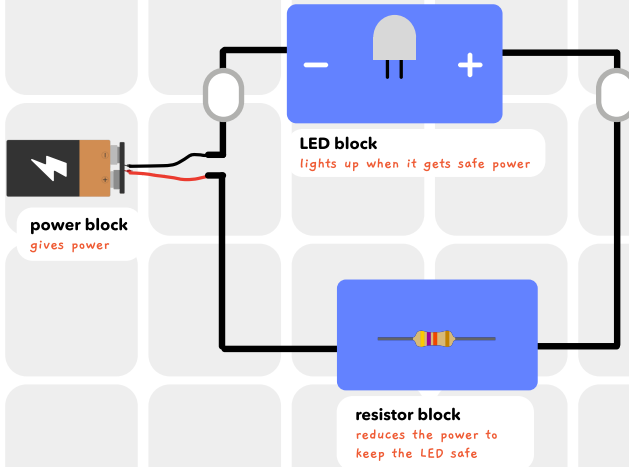


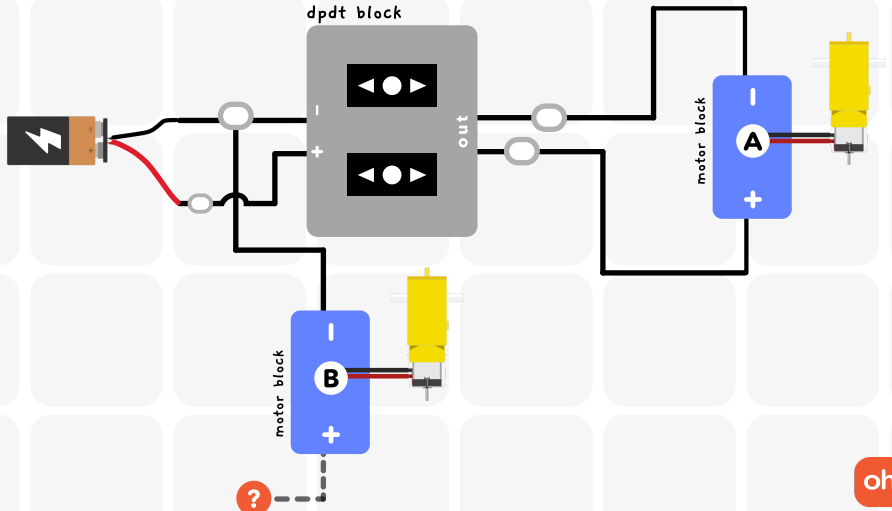
circuit 1 : resistor block & led block

Task: Connect all blocks as per the circuit below such that the LED lights up when it gets safe power.



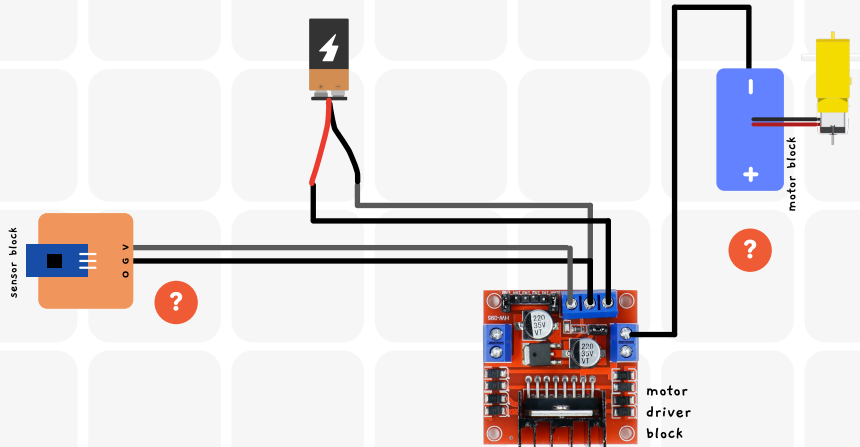
circuit 1: dpdt block + motor block

Task: Control motor block (A) using a dpdt block while motor block (B) should remain ON. Can you complete the missing wire ? in the circuit?



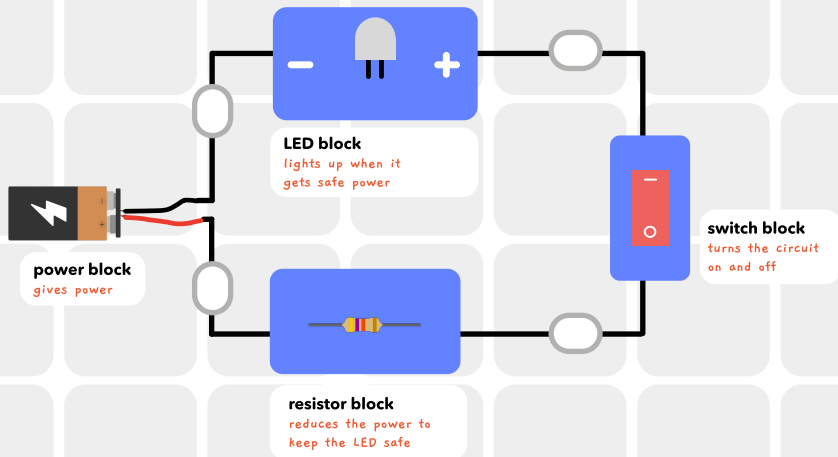
circuit 1: ir sensor + motor block

Task: Use a motor driver to control a motor block. Control the direction of motor by swapping the Out PIN of the sensor with IN1 and IN2 of the motor driver. Can you complete the missing wire **?** in the circuit and change the direction of the motor using the sensor?



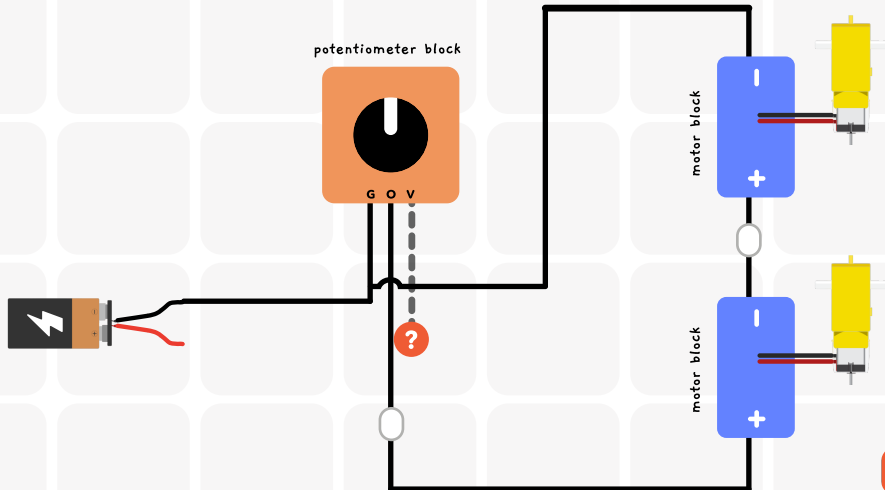
circuit 2 : switch block

Task: Connect all blocks in a circuit and place the switch such that the LED turns on when the switch is ON.



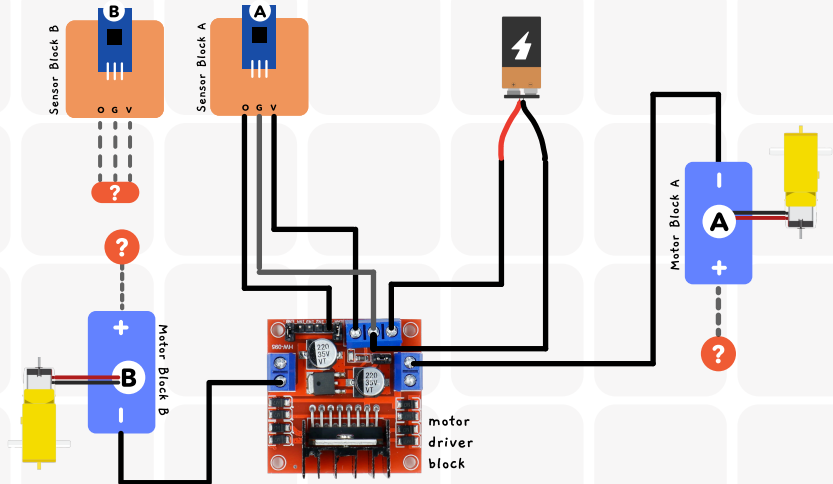
circuit 2: potentiometer + dual motor blocks

Task: A circuit uses a pot block to control the speed of 2 motor blocks connected in series. Can you complete missing wire **?** in the circuit and adjust the speed?



circuit 2: dual ir sensor circuit

Task: Use a motor driver to control 2 motor blocks. The signal to the motor block (A) is sent through the IR sensor block(A) and the signal to the motor block (B) is sent through the IR sensor block(B). Can you complete the missing wire **?** in the circuit and control the motors using the sensors?

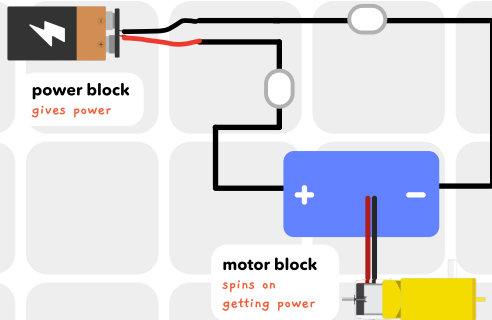


circuit 3 : motor block

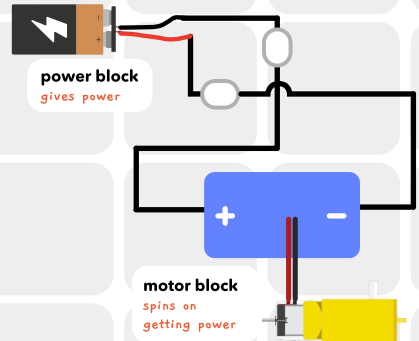
Task A: Connect the battery to the motor and observe the motor spin.

Task B: Reverse the wire connections and observe how the motor spins in the opposite direction.

a.

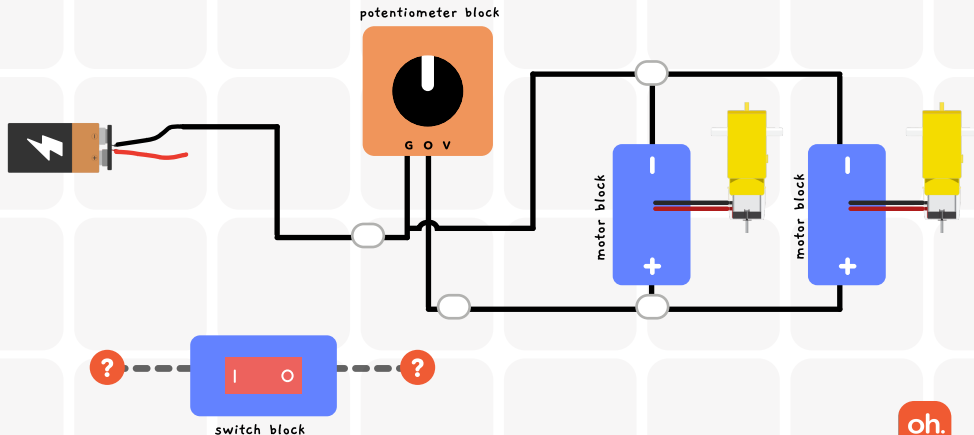


b.



circuit 3 : potentiometer + switch block

Task: Use a potentiometer block to control the speed of 2 motor blocks connected in parallel. The circuit needs to be turned ON/OFF using a switch block. Can you complete missing wire **?** in the circuit and adjust the speed?

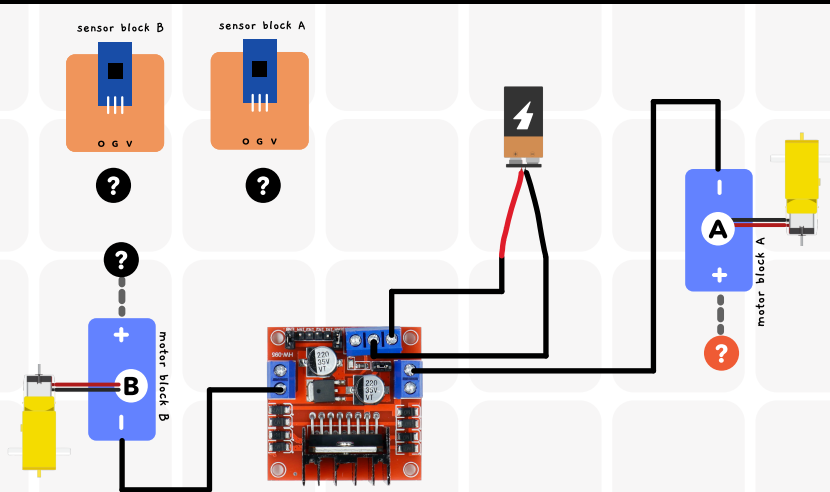


circuit 3: dual motor blocks + ir sensors

Task: Use a motor driver to control 2 motor blocks.

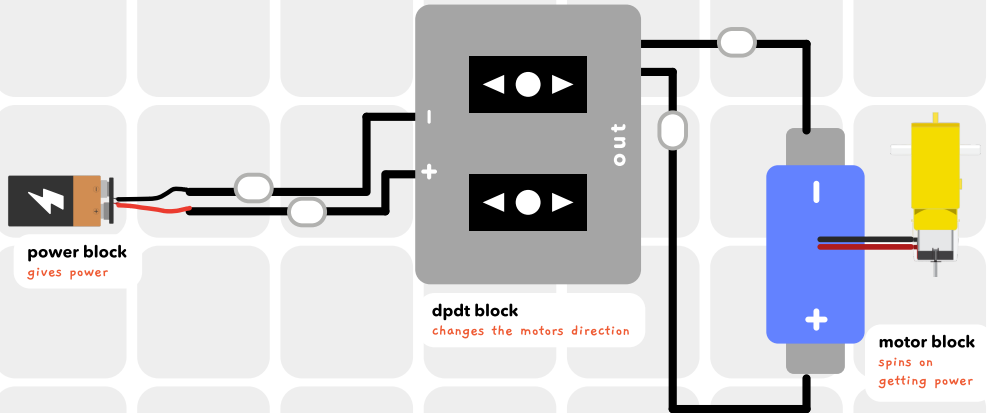
Motor block (A) must turn Right and Motor block (b) must turn left.

Use the Sensor OUT pins to control the direction of the motors and complete the missing wires **?** in the circuit.



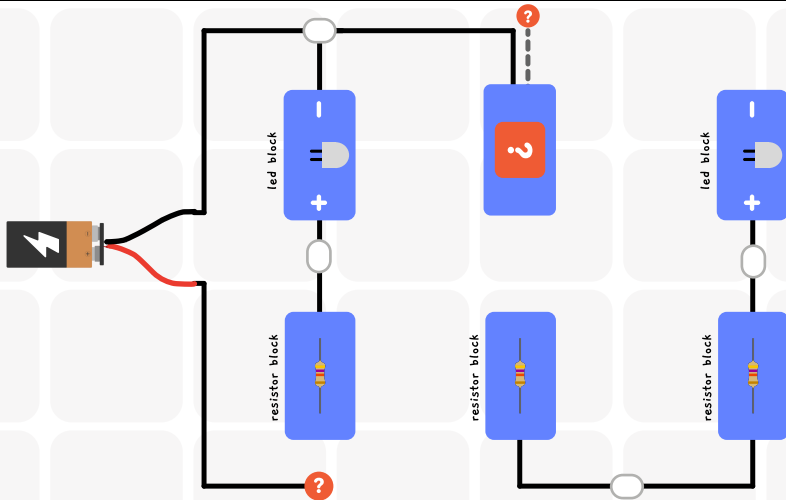
circuit 4 : dpdt block

Task: Connect the battery and motor through the DPDT block.
Press the DPDT switch to change the direction of the motor.



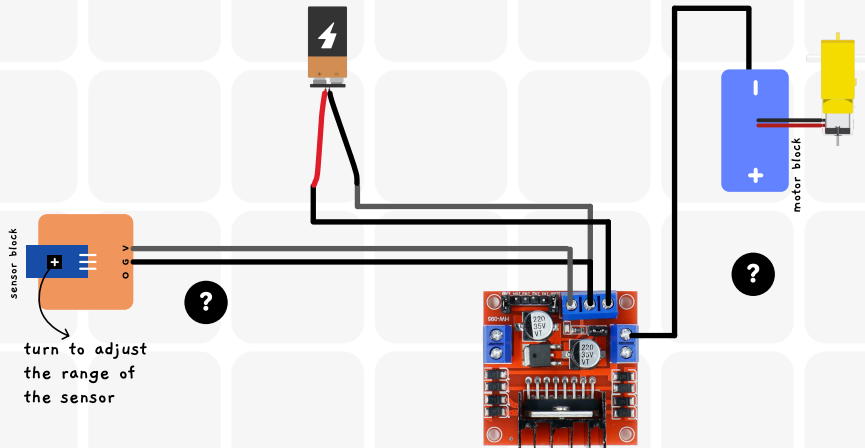
circuit 4 : parallel circuit

Task: Make a parallel circuit with the following and find the missing block  and missing wires  to complete the circuit.



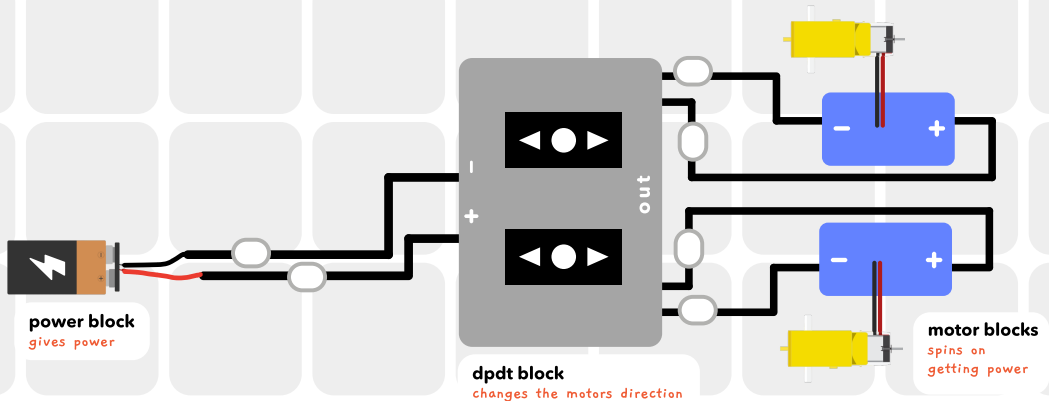
circuit 4: ir sensor + motor block

Task: Send a signal to the motor driver block using an IR sensor block. Adjust the range of the IR Sensor to detect objects that are close enough. Complete the missing wires **?** in the circuit.



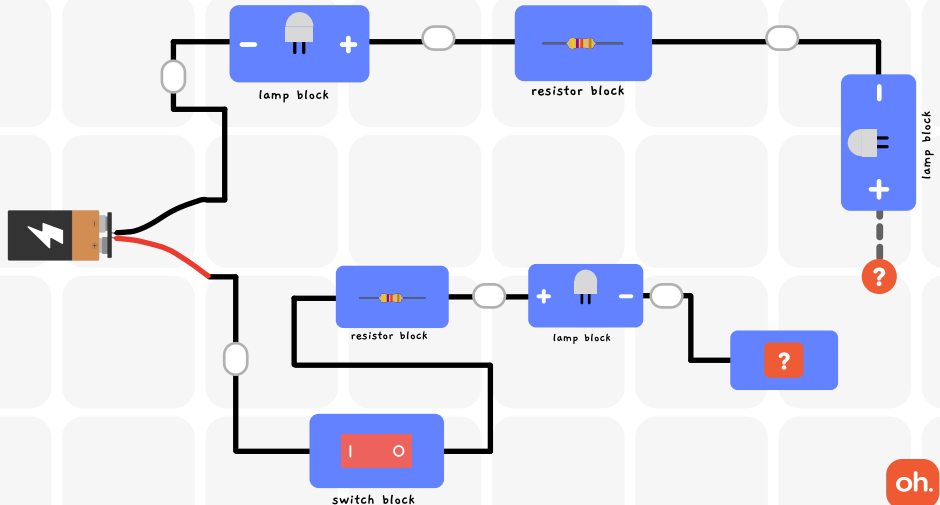
circuit 5 : dual dpdt circuit

Task: Connect each motor to its own DPDT switch and to the battery.
Press each DPDT switch to control each motor's direction.



circuit 5 : series circuit

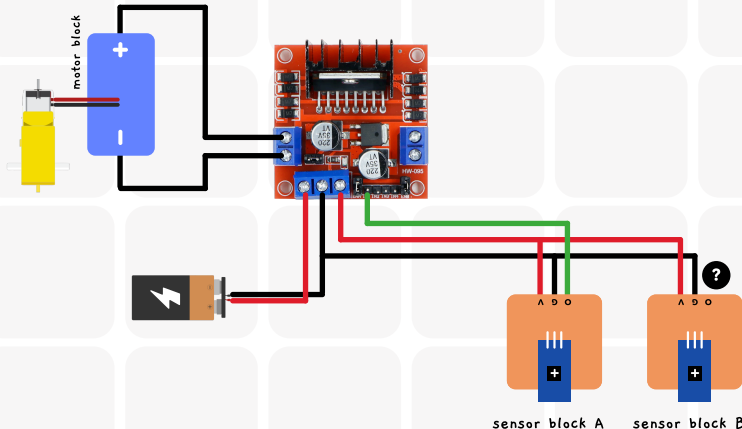
Task: Make a series circuit with the following and find the missing block ?
and missing wire ? to complete the circuit.



circuit 5: dual ir sensor + motor

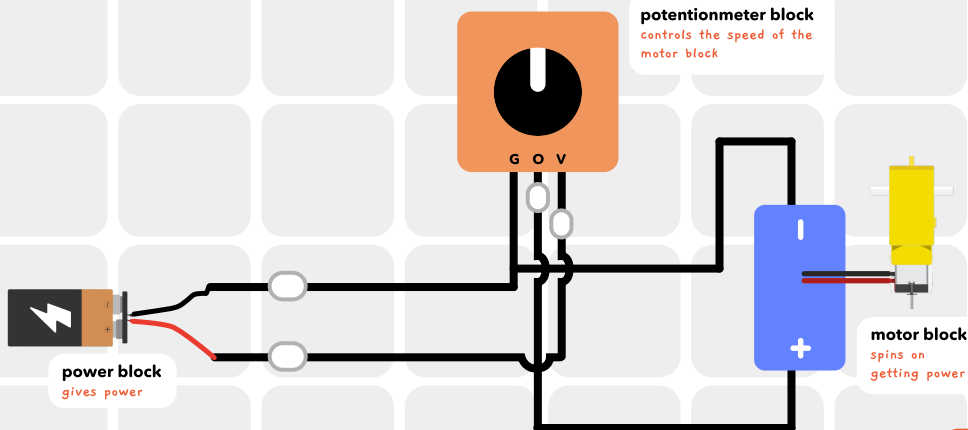
Task: Use a motor driver to control a motor block.

The signal to the motor block is sent through the IR Sensor block (A) to move forward and IR sensor block (B) to move backward. Can you complete the missing wire in the circuit and control the motors using the sensors?



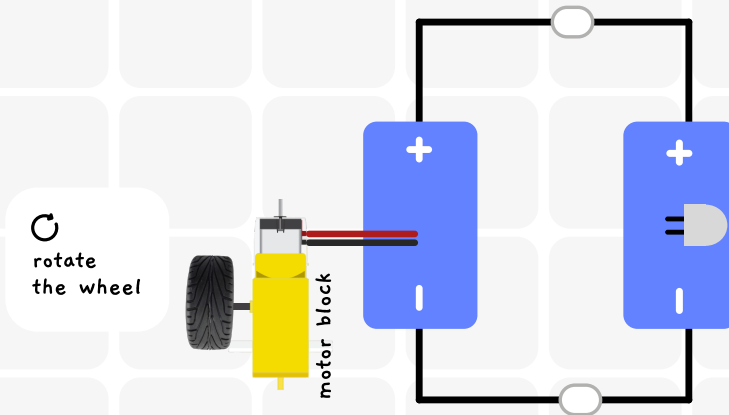
circuit 6 : potentiometer block

Task: Connect the battery and motor through the potentiometer block.
Turn the knob to control the speed of the motor.



circuit 6: motor + led

Task: Connect the Motor to the LED.
Rotate the wheel to light up the LED!

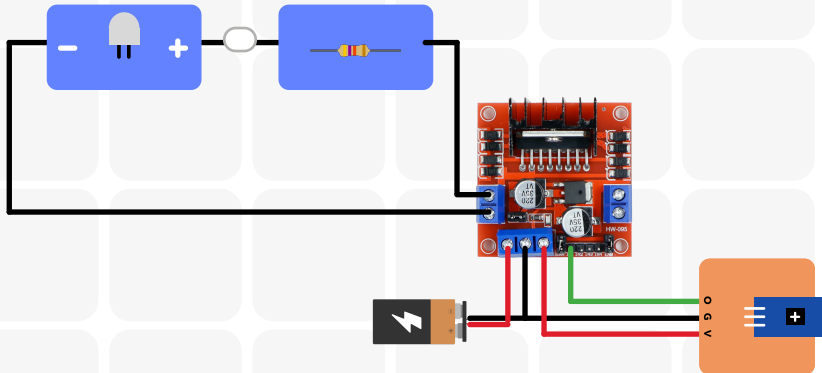


circuit 6:

Ldr sensor + motor driver + led & resistor

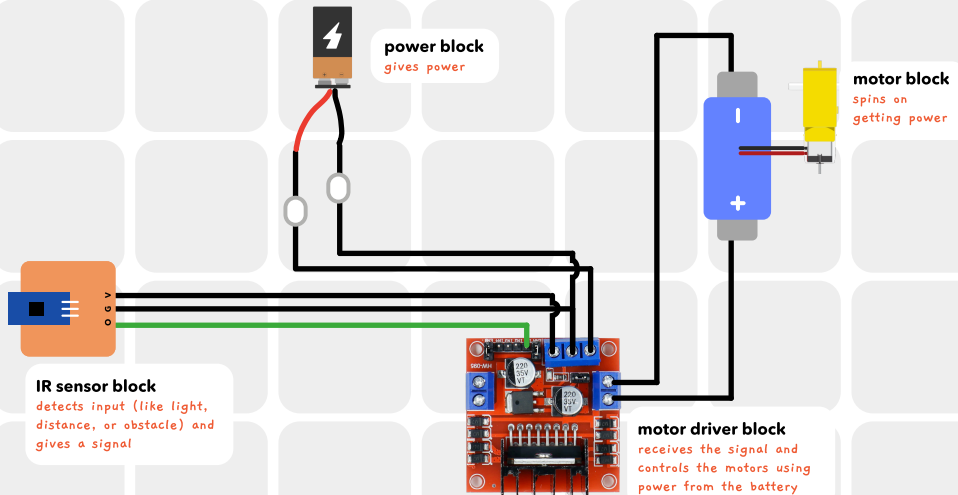
Task: Connect the LED to the Motor Driver.

The input signal to the motor driver is sent using a LDR Sensor to turn LED ON/OFF.



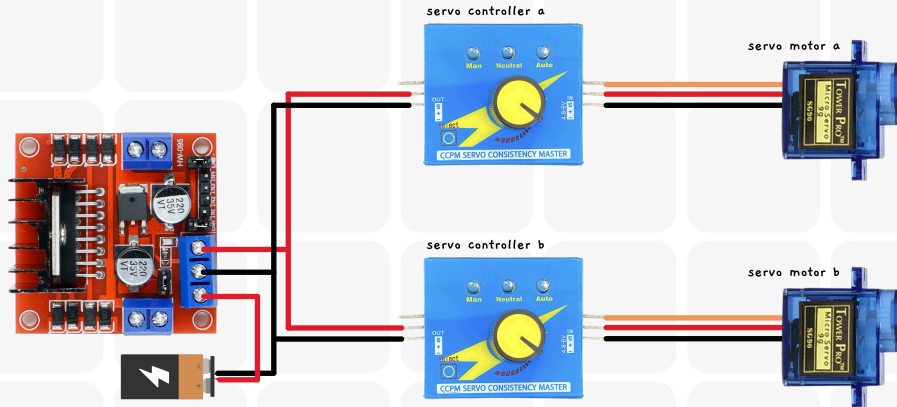
circuit 7 : motor driver + ir sensor

Task: Connect the IR sensor to the motor driver and connect the motor to the driver outputs. The IR sensor's signal will tell the motor driver when to run the motor, such that the motor moves only when the IR sensor is activated.



circuit 7: servo motor + servo controller

Task: Control servo motor(a) using Servo controller(a) in manual mode and Servo motor (b) using servo controller(b) in Auto mode.



circuit 7:

ldr sensor + ir sensor + motor driver + led & resistor

Task: Connect the LED to the Motor Driver.

The input signal to the motor driver is sent using a LDR Sensor to turn LED ON/OFF.

